Pro Football Sports Science Research Update

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3-Dimensional biomechanics of noncontact anterior cruciate ligament injuries in male professional soccer players. The American Journal of Sports Medicine, Ahead of Print: 1-10, 2024.

A review of elite athlete evidence-based knowledge and preparation for competing in the heat. Journal of Science in Sport and Exercise, Ahead of Print: 1-20, 2024.

Acute:chronic workload ratio and training monotony variations over the season in professional soccer: A systematic review. Journal of Sports Engineering and Technology, Ahead of Print: 1-16, 2023.

Addressing muscle-tendon imbalances in adult male athletes with personalized exercise prescription based on tendon strain. European Journal of Applied Physiology, Ahead of Print: 1-14, 2024.

Avoiding the edge before it its too late: When is pain a sign of injury in athletes? Clinical Journal of Sport Medicine, Ahead of Print: 1-3, 2024.

Balancing the load: A narrative review with methodological implications of compensatory training strategies for non-starting soccer players. Biology of Sport, 41(4): 173-185, 2024.

Can improving postoperative sleep speed up surgical recovery? Sleep Science, Ahead of Print: 1-4, 2024.

Cardiac cycle timing and contractility following acute sport-related concussion. Research in Sports Medicine, 32(2): 260-267, 2024.

Concerns about blood clots in sports medicine. Current Sports Medicine Reports, 23(5): 159-160, 2024. (No abstract available)

Concussion in the sleeping brain. Sports Medicine – Open, 10:68, 2024.

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Contaminants in dietary supplements: Toxicity, doping risk, and current regulation. International Journal of Sport Nutrition and Exercise Metabolism, Ahead of Print: 1-10, 2024.

Current evidence on common dietary supplements for sleep quality. American Journal of Lifestyle Medicine, 18(3): 323-327, 2024.

Different anatomic patterns of the indirect tendon of the rectus femoris. Surgical and Radiologic Anatomy, Ahead of Print: 1-8, 2024.

Distal tibiofibular syndesmosis injuries in the National Football League (NFL): A spectrum of pathology that correlates with time to return to full participation. Sports Health, Ahead of Print: 1-8, 2024.

Dynamic visual acuity, vestibulo-ocular reflex, and visual field in National Football League (NFL) officiating: Physiology and visualization engineering for 3D virtual on-field training. Vision, 8:35, 2024.

Effects of photobiomodulation on pain and return to play of injured athletes: A systematic review and meta-analysis. Journal of Strength and Conditioning Research, 38(6): e310-e319, 2024.

Effects of recent cannabis consumption on eye-tracking and pupillometry. Frontiers in Neuroscience, 18:1358491, 2024.

Elevated brain temperature under severe heat exposure impairs cortical motor activity and executive function. Journal of Sport and Health Science, 1392): 233-244, 2024.

Elite athletes' and support staffs' experiences and perceptions of long-haul travel, and the self-management strategies they use. International Journal of Sports Science & Coaching, 19(2): 1-12, 2024.

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Epidemiology of hamstring tears in NCAA athletes: Findings from the NCAA Injury Surveillance Program between 2014/2015 and 2018/2019. Clinical Journal of Sport Medicine, Ahead of Print: 1-10, 2024.

Head acceleration event exposure and cognitive and functional outcomes: A comparison of multiple football seasons. Research in Sports Medicine, 32(1): 122-131, 2024.

How mental health affects injury risk and outcomes in athletes. Sports Health, Ahead of Print: 1-8, 2024.

Human foot muscle strength and its association with sprint acceleration, cutting and jumping performance, and kinetics in high-level athletes. Journal of Sports Sciences, Ahead of Print: 1-12, 2024.

Influence of the external workload on calf muscle strain injuries in professional football players: A pilot study. Sports Health, Ahead of Print: 1-8, 2024.

Knee cartilage injuries in football players: Clinical outcomes and return to sport after surgical treatment: A systematic review of the literature. Cartilage, Ahead of Print: 1-15, 2024.

Long-term outcomes of multiligament knee injuries in American football players. The American Journal of Sports Medicine, Ahead of Print: 1-9, 2024.

Maximum speed exposures in Australian Rules football: Do methods matter? Science and Medicine in Football, 8(3): 287-290, 2024.

NFL quarterbacks with ulnar collateral ligament injuries have high return to play rates, but older players have inferior post-injury performance. Arthroscopy, Sports Medicine, and Rehabilitation, Ahead of Print: 1-30, 2024.

Overview of treatment options for mild traumatic brain injury: A literature review. Cureus, 16(4): e59021, 2024.

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Physician perceptions of cannabidiol (CBD) and cannabis in sports medicine and performance. Translational Sports Medicine, 11:2023:8824466, 2023.

Predictors of early return to sport after surgical repair of proximal hamstring complex injuries in professional athletes: A prospective study. The American Journal of Sports Medicine, 52(4): 1005-1013, 2024.

Preventing exertional heat stroke in football: Time for a paradigm shift. Sports Health, Ahead of Print: 1-7, 2024.

Reasons for not returning to pre-injury sport level after ACL-reconstruction. International Journal of Sports Medicine, Ahead of Print: 1-7, 2024.

Return to the pre-injury level of sport after anterior cruciate ligament reconstruction: A practical review with medical recommendations. International Journal of Sports Medicine, Ahead of Print: 1-17, 2024.

Skimmed, lactose-free milk ingestion postexercise: Rehydration effectiveness and gastrointestinal disturbances vs water and a sports drink in physically active people. International Journal of Sport Nutrition and Exercise Metabolism, Ahead of Print: 1-9, 2024.

Subconcussive is a dangerous misnomer: Hits of greater magnitude than concussive impacts may not cause symptoms. British Journal of Sports Medicine, Ahead of Print: 1-3, 2024. (No abstract available)

TeamWellX: Leveraging genetic algorithms to optimize team health and well-being toward sustainable game development and strategy. SoftwareX, 101635, 2024.

The effect of variations in knee and hip angles on electromyographic activity of the hamstrings and related muscles during the Nordic hamstring exercise. International Journal of Sports Physical Therapy, 19(3): 275-283, 2024.

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Thoracolumbar fascia and lumbar muscle stiffness in athletes with a history of hamstring injury. Journal of Sports Science and Medicine, 23: 436-444, 2024.

Validity of the Hawkin Dynamics wireless dual force platform system against a piezoelectric laboratory grade system for vertical countermovement jump variables. Journal of Strength and Conditioning Research, 38(6): 1144-1148, 2024.

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3-Dimensional biomechanics of noncontact anterior cruciate ligament injuries in male professional soccer players. The American Journal of Sports Medicine, Ahead of Print: 1-10, 2024.

Background: The understanding of noncontact anterior cruciate ligament (ACL) injury causation in soccer has improved over the past decades. Bidimensional video analyses have significantly augmented our awareness, representing to date the only practical method to describe injury biomechanics. However, the extent of the problem continues to raise serious concerns.

Purpose: To advance our understanding of the causal pathways leading to ACL injury with a large-scale reconstruction of 3-dimensional (3D) whole-body joint kinematics of injuries that occurred to male elite soccer players, as well as to compare the joint angle time course among situational patterns.

Study design: Descriptive laboratory study.

Methods: A total of 33 consecutive noncontact and indirect contact ACL injuries that occurred in 6 national and 2 international professional leagues (seasons 2020-2021 to 2022-2023 until December 2022) were analyzed: (1) multiview noncoaxial television images were inspected; (2) multiple camera views were taken from 400 ms before the initial ground contact to 200 ms after the injury frame; (3) a size-matched pitch was modeled and used to calibrate cameras; (4) a 3D skeletal model was adjusted to fit the player's pose in each frame/view; and (5) poses were interpolated, and Euler joint angles were extracted.

Results: The authors reconstructed the 3D lower limb joint kinematic curves preceding and during ACL injuries in 33 cases; notably, a sudden external (up to 5°) and then internal knee rotation was observed after the initial contact and before the injury frame. The overall kinematics at injury were knee moderately flexed (45.9°± 21.7°), abducted (4.3°± 5.1°), and externally rotated (3.0°± 6.4°); trunk shallowly flexed (17.4°± 12.5°) and rotated and tilted toward the injured side; and hip flexed (32.0°± 18.7°), abducted (31.1°± 12.0°), and slightly internally rotated (6.6°± 12.2°). Variable behaviors were observed at the ankle level.

Conclusion: Via reconstruction of the sequence of whole-body joint motion leading to injury, we confirmed the accepted gross biomechanics (dynamic valgus trend). This study significantly enriches the

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current knowledge on multiplanar kinematic features (transverse and coronal plane rotations). Furthermore, it was shown that ACL injuries in male professional soccer players manifest through distinct biomechanical footprints related to the concurrent game situation.

Clinical relevance: Interventions aimed at reducing ACL injuries in soccer should consider that environmental features (ie, situational patterns) affect injury mechanics.

A review of elite athlete evidence-based knowledge and preparation for competing in the heat. Journal of Science in Sport and Exercise, Ahead of Print: 1-20, 2024.

Abstract - Elite athletes will compete in extreme heat more frequently as global land and sea temperatures increase, alongside more intense, frequent and longer duration heatwaves. Best practices to protect athlete health and performance during competition include heat acclimation/acclimatization [(HA); i.e., long-term pre-competition preparation] complemented by pre-planned and practiced cooling and hydration strategies (i.e., short-term interventions immediately before or during competition). This review explores elite athletes' current behaviors and practices when preparing for competition in the heat and assesses the level of knowledge that has been exhibited by athletes and their practitioners in this space. Recommendations for future research, discussions of current best practices, and methods to improve translation of research into practice are provided. Available research focuses on small samples of elite endurance athletes during a selection of World Championship/Olympic/ Paralympic events (~6% of competing athletes). While generally an increase in the adoption of evidence-based HA is seen chronologically from 2015 onwards, universal adoption is not seen. HA adoption is lowest in those who live/train in cold/ temperate environments with cost and access to facilities/equipment being the most commonly reported barriers. Further research is required across the sporting landscape to fully characterize elite athlete behaviors and practices in these spaces. International federations and national governing bodies should continue their efforts to educate athletes and focus on regularly updated and reinvigorated release of evidence-based guidelines (in multiple germane languages) for competing in the heat, to increase the adoption of HA and other heat related best practice.

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Acute:chronic workload ratio and training monotony variations over the season in professional soccer: A systematic review. Journal of Sports Engineering and Technology, Ahead of Print: 1-16, 2023.

Abstract - Acute: chronic workload ratio (ACWR) and training monotony have been criticized as injury risk predictors. Therefore, the use of intensity measures should be oriented to understand the variations of intensity across the season. The aim of this systematic review is to summarize the main evidence about the ACWR and training monotony variations over the season in professional soccer players. The search was made in PubMed, SPORTDiscus, and FECYT according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines. From the 225 studies initially identified, 27 were fully reviewed, and their outcome measures were extracted and analyzed. Existing literature revealed a variety of designs, ACWR and training monotony ranges, variables assessed and durations of the studies. Overall, the range values for ACWR were 0.4–3.39 AU, while those focused on monotony were 0.49–5.7 AU. Regarding ACWR, the ratios located around 0.85–1.25 could predict lower risk values and ratios around ≥ 1.50 could predict higher risk values. On the contrary, with respect to training monotony, the ratios are approximately between 0.5 and 2.00 (low values in the preseason and low competition weeks and high values when soccer players are in highly scheduled competition weeks). Nevertheless, ACWR and training monotony methods should be addressed and considered based on their real value before using this indicator to reduce injury risk. In fact, the data did not conclusively define injured and non-injured players. For this reason, utilizing standardized approaches will allow for more precise conclusions about professional soccer players.

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Addressing muscle-tendon imbalances in adult male athletes with personalized exercise prescription based on tendon strain. European Journal of Applied Physiology, Ahead of Print: 1-14, 2024.

Purpose: Imbalances of muscle strength and tendon stiffness can increase the operating strain of tendons and risk of injury. Here, we used a new approach to identify muscle-tendon imbalances and personalize exercise prescription based on tendon strain during maximum voluntary contractions (εmax) to mitigate musculotendinous imbalances in male adult volleyball athletes.

Methods: Four times over a season, we measured knee extensor strength and patellar tendon mechanical properties using dynamometry and ultrasonography. Tendon micromorphology was evaluated through an ultrasound peak spatial frequency (PSF) analysis. While a control group (n = 12) continued their regular training, an intervention group (n = 10) performed exercises (3 × /week) with personalized loads to elicit tendon strains that promote tendon adaptation (i.e., 4.5-6.5%).

Results: Based on a linear mixed model, εmax increased significantly in the control group over the 9 months of observation (pCon = 0.010), while there was no systematic change in the intervention group (pInt = 0.575). The model residuals of εmax, as a measure of imbalances in muscle-tendon adaptation, demonstrated a significant reduction over time exclusively in the intervention group (pInt = 0.007). While knee extensor muscle strength increased in both groups by ~ 8% (pCon < 0.001, pInt = 0.064), only the intervention group showed a trend toward increased normalized tendon stiffness (pCon = 0.824, pInt = 0.051). PSF values did not change significantly in either group (p > 0.05).

Conclusion: These results suggest that personalized exercise prescription can reduce muscle-tendon imbalances in athletes and could provide new opportunities for tendon injury prevention.

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Avoiding the edge before it its too late: When is pain a sign of injury in athletes? Clinical Journal of Sport Medicine, Ahead of Print: 1-3, 2024.

Introduction - A multitude of internal and external factors converge to produce the output of pain. Although pain is an anticipated side effect of high-intensity exercise, if athletes persist through injury-related pain mistaken for muscle soreness, it may "push them over the edge," leading to injury. The decision to push through or pull back ultimately lies with the athlete. Although athletes are often adept at making this decision, even at the highest level their decision-making skills are not infallible, with approximately 9% of athletes at the 2022 Summer Olympics sustaining an injury. Athletes typically learn the limits of their personal edge and the difference between pain that is a sign of training to one's limit versus injury based on trial-and-error in their decision making. To increase precision in this process, enhancement in the understanding of the meaning of pain (eg, threat value) from the cells to the cortex and how to respond optimally in a given context is required. As a complex array of factors contribute to defining this edge, advancement in precisely targeted, multimodal biopsychosocial assessment has the potential to isolate individualized early warning signs.

Balancing the load: A narrative review with methodological implications of compensatory training strategies for non-starting soccer players. Biology of Sport, 41(4): 173-185, 2024.

Abstract - New training approaches have emerged advocating for the implementation of compensatory physical training. This approach aims to provide additional training that balances the load typically experienced by non-starters during a match. This may help maintain their readiness and ensures that their physical fitness is not compromised by the reduced exposure to match loads. Thus, this narrative review aims to describe the differences in external loads between starting and non-starting players and describe the studies conducted in compensatory training. Studies examining external load metrics such as total distances covered, high-speed running, and sprinting suggest that, adjusted for playing time, values are often higher in non-starting players. Although not standardized, there is an obvious decrease in exposure for these critical variables in non-starters. Additionally, internal load parameters such as perceived exertion and heart rate tend to be higher in starting players. Regarding the physical fitness

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impacts, evidence suggests differences observed between starters and non-starters in some aspects of physical performance, although the extent and significance of these differences can vary. The studies on compensatory training are limited, and the typical approach usually centers on running-based exercises and small-sided games, offering differing approaches to address the physical needs. The gap in research underscores the necessity for improved study designs that can shed light on the real impact of compensatory training. Presently, the practice of compensatory training has been adopted, yet a definitive understanding of its genuine influence, particularly in terms of enhancing physical fitness and mitigating injury risks, remains elusive.

Can improving postoperative sleep speed up surgical recovery? Sleep Science, Ahead of Print: 1-4, 2024.

Abstract - Sleep disturbance is common during recovery after surgical procedures and may have an important effect on mortality, and quality of life. Sleep restriction/deprivation, including decreased quantity and continuity, is common in patients who are patients and persons with acute and chronic illnesses. Age, gender, illness, primary sleep disorders, environment, and medical treatment factors are thought to influence sleep throughout the preoperative period, hospitalization, and recovery. Resulting sleep pattern disturbances include decreases in circadian patterning, continuity, duration, and perceived (subjective) sleep quality. This article synthesizes sleep disturbance in patients who have undergone surgery and highlights sleep strategies to improve faster surgical recovery.

Cardiac cycle timing and contractility following acute sport-related concussion. Research in Sports Medicine, 32(2): 260-267, 2024.

Abstract - Cardiac sequelae following sport-related concussion are not well understood. This study describes changes in the cardiac cycle timing intervals and contractility parameters during the acute phase of concussion. Twelve athletes (21 ± 2 years, height = 182 ± 9 cm, mass = 86 ± 15 kg, BMI = 26 ± 3 kg/m2) were assessed within 5 days of sustaining a diagnosed concussion against their own pre-season baseline. A non-invasive cardiac sensor (LLA RecordisTM) was used to record the cardiac cycle

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parameters of the heart for 1 minute during supine rest. Cardiac cycle timing intervals (Isovolumic relaxation and contraction time, Mitral valve open to E wave, Rapid ejection period, Atrial systole to mitral valve closure, Systole, and Diastole) and contractile forces (Twist force and Atrial systole: AS) were compared. Systolic time significantly decreased during acute concussion ($p = 0.034$). Magnitude of AS significantly increased during acute concussion ($p = 0.013$). These results imply that concussion can result in altered systolic function.

Concussion in the sleeping brain. Sports Medicine – Open, 10:68, 2024.

Background: Emerging research has suggested sleep to be a modifier of the trajectory of concussion recovery in adolescent and adult populations. Despite the growing recognition of the relationship between sleep and concussion, the mechanisms and physiological processes governing this association have yet to be established.

Main body: Following a concussion, a pathophysiologic cascade of events occurs, characterized by numerous factors including microglia activation, ionic imbalance, and release of excitatory neurotransmitters. Importantly, each of these factors plays a role in the regulation of the sleep-wake cycle. Therefore, dysregulation of sleep following injury may be a function of the diffuse disruption of cerebral functioning in the wake of both axonal damage and secondary physiological events. As the onset of sleep-related symptoms is highly variable following a concussion, clinicians should be aware of when and how these symptoms present. Post-injury changes in sleep have been reported in the acute, sub-acute, and chronic phases of recovery and can prolong symptom resolution, affect neurocognitive performance, and influence mood state. Though these changes support sleep as a modifier of recovery, limited guidance exists for clinicians or their patients in the management of sleep after concussion. This may be attributed to the fact that research has correlated sleep with concussion recovery but has failed to explain why the correlation exists. Sleep is a complex, multifactorial process and the changes seen in sleep that are seen following concussion are the result of interactions amongst numerous processes that regulate the sleep-wake cycle.

Short conclusion: The assessment and management of sleep by identifying and considering the biological, sociological, and psychological interactions of this multifactorial process will allow for clinicians to address the dynamic nature of changes in sleep following concussion.

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Contaminants in dietary supplements: Toxicity, doping risk, and current regulation. International Journal of Sport Nutrition and Exercise Metabolism, Ahead of Print: 1-10, 2024.

Abstract - Athletes, both amateur and professional, often resort to the consumption of nutritional supplements without professional supervision and without being aware of the risks they may entail. We conducted an exhaustive literature search to determine the most common substances found as contaminants in dietary supplements. For each substance, we analyzed its mechanism of action, clinical indication, health risk, and putative use as doping agent. In addition, we evaluated the current regulation of these supplements. Contamination of nutritional supplements (accidental or intentional), especially with steroids and stimulants, is a hazardous situation. The prolonged consumption of these products without being aware of their composition can cause serious health risks and, in the case of professional athletes, a possible sanction for doping.

Current evidence on common dietary supplements for sleep quality. American Journal of Lifestyle Medicine, 18(3): 323-327, 2024.

Abstract - Poor sleep quality impacts nearly 70 million adults in the United States, resulting in nearly 1 in 5 adults regularly utilizing some sleep aid, either prescription or over-the-counter. Common dietary supplements utilized to improve sleep quality include nitrates, melatonin, magnesium, zinc, vitamin D, and L-theanine. While underlying physiologic mechanisms support the potential impact these compounds have on sleep quality, evidence from clinical trials varies widely. Melatonin and magnesium are two of the more widely researched supplements, which have numerous studies, with conflicting results. Other compounds such as nitrates, zinc, vitamin D, and L-theanine are less supported. Additional research is needed in order to more appropriately recommend these dietary supplements for sleep improvements.

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Different anatomic patterns of the indirect tendon of the rectus femoris. Surgical and Radiologic Anatomy, Ahead of Print: 1-8, 2024.

Purpose: The rectus femoris forms the anterior portion of the quadriceps muscle. It has a proximal tendinous complex, which is constituted by a direct tendon, an indirect tendon, and a variable third tendon. Direct and indirect tendons converge into a common tendon. The purposes of this study are to add anatomical knowledge about the proximal tendinous complex and describe anatomical variants of the indirect tendon and, on these basis, categorize different anatomical patterns.

Method: In this study, 48 hemipelvis from bodies donated to the Universitat Autònoma de Barcelona have been dissected to examine the proximal tendinous complex of the rectus femoris.

Results: The following anatomical variants of the indirect tendon were described: inferior aponeurotic expansion in 23/48 cases (47.9%); superior aponeurotic expansion in 21/48 cases (43.7%); and an unusual origin of the myotendinous junction of the rectus femoris in the free portion of the indirect tendon in 19/48 cases (39.6%). On the basis of the aponeurotic expansions, the following anatomical patterns of the indirect tendon were defined: standard (19/48 cases, 39.6%), superior and inferior complex (15/48 cases, 31.2%), inferior complex (8/48 cases, 16.7%), and superior complex (6/48 cases, 12.5%).

Conclusion: We can categorize four different anatomical patterns of the indirect tendon, three of which are complex. We suggest that complex patterns can cause an increased stiffness of the indirect tendon and so be considered non-modifiable risk factors for rectus femoris injuries. Finally, it would be useful to identify complex patterns and perform injury prevention actions through specific physical preparation programs.

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Distal tibiofibular syndesmosis injuries in the National Football League (NFL): A spectrum of pathology that correlates with time to return to full participation. Sports Health, Ahead of Print: 1-8, 2024.

Background: Studies routinely evaluate high ankle sprains in isolation, but recent data suggest that these injuries are often associated with concomitant pathology, potentially influencing return to full participation.

Hypothesis: In National Football League (NFL) players, isolated high ankle sprains are rare and syndesmosis injuries with concomitant pathology will result in increased time to return to full participation.

Study design: Cross-sectional study.

Level of evidence: Level 3.

Methods: Distal tibiofibular syndesmosis injuries sustained by NFL players between 2017 and 2019 were identified through NFL Injury Surveillance Database queries and verified with video analysis. Each injury underwent a comprehensive magnetic resonance imaging (MRI) review. Regression modeling was utilized to explore the influence of numerous imaging findings on time to return to full participation.

Results: There were 83 external rotation ankle injuries involving the syndesmosis. Isolated distal tibiofibular syndesmosis injuries were rare (n = 11; 13%) and more often associated with other ligamentous injury (deltoid ligament and lateral ligamentous complex) and/or fractures. Regression modeling resulted in clustering of 3 injury pattern groups for time to return to full participation across numerous imaging findings: syndesmosis injury-fracture combinations (250 days [interquartile range [IQR,] 142-266 days]), syndesmosis injury with complete deep deltoid tear or acute diastasis (175 days [IQR, 20-248 days]), and all other syndesmosis injuries (27 days [IQR, 18-46 days]).

Conclusion: In NFL athletes with external rotation ankle injuries, isolated distal tibiofibular syndesmosis injuries were rare and more often associated with concomitant pathology. Time to return to full participation was affected by an associated fracture and complete deep deltoid ligament tear or diastasis but no other relevant MRI variables such as lateral ligament complex involvement or the presence of osteochondral lesions or bone contusions.

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Dynamic visual acuity, vestibulo-ocular reflex, and visual field in National Football League (NFL) officiating: Physiology and visualization engineering for 3D virtual on-field training. Vision, 8:35, 2024.

Abstract - The ability to make on-field, split-second decisions is critical for National Football League (NFL) game officials. Multiple principles in visual function are critical for accuracy and precision of these play calls, including foveation time and unobstructed line of sight, static visual acuity, dynamic visual acuity, vestibulo-ocular reflex, and sufficient visual field. Prior research has shown that a standardized curriculum in these neuro-ophthalmic principles have demonstrated validity and self-rated improvements in understanding, confidence, and likelihood of future utilization by NFL game officials to maximize visual performance during officiating. Virtual reality technology may also be able to help optimize understandings of specific neuro-ophthalmic principles and simulate real-life gameplay. Personal communication between authors and NFL officials and leadership have indicated that there is high interest in 3D virtual on-field training for NFL officiating. In this manuscript, we review the current and past research in this space regarding a neuro-ophthalmic curriculum for NFL officials. We then provide an overview our current visualization engineering process in taking real-life NFL gameplay 2D data and creating 3D environments for virtual reality gameplay training for football officials to practice plays that highlight neuro-ophthalmic principles. We then review in-depth the physiology behind these principles and discuss strategies to implement these principles into virtual reality for football officiating.

Effects of photobiomodulation on pain and return to play of injured athletes: A systematic review and meta-analysis. Journal of Strength and Conditioning Research, 38(6): e310-e319, 2024.

Abstract - The aims of this systematic review and meta-analysis were to evaluate the effect of photobiomodulation (PBM) on musculoskeletal pain in injured athletes and to determine if the effects of PBM allowed injured athletes to return to play faster. Electronic databases (MEDLINE Complete, CINAHL, and SPORTDiscus, PubMed, Web of Science, and Embase) were systematically searched (up to and including November 7, 2023) for peer-reviewed randomized controlled trials (RCTs) meeting criteria. Six RCTs, representing 205 competitive and recreational athletes with a mean age of 24 years, were

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included in the analysis. There were 6 intervention groups using standard physical therapy (n = 1), placebo PBM (n = 4), and aloe gel (n = 1) lasting between 10 minutes and 8 weeks in duration. The level of significance set for the study was $p < 0.05$. Overall, the use of PBM indicated a positive effect on pain reduction for PBM vs. control groups, standardized mean differences = 1.03, SE 5 0.22, 95% confidence intervals = [0.43–1.63], p = 0.0089, but the 2 RCTs found evaluating the effect of PBM on time to return to play after injury in athletes do not support a benefit. Allied healthcare professionals may use PBM to reduce pain, thus allowing an athlete to return to their normal biomechanical movement faster; however, limited evidence suggests that PBM does not reduce time to return to play after an injury.

Practical Applications - Our meta-analysis indicates that PBM can reduce pain in athletes after injury. Those responsible for acquiring equipment should select devices that emit wavelengths between 810 and 980 nm because these were used in the included studies. Commercially available PBM handheld devices range in price from $150 to $30,0000 USD, with whole-body PBM equipment (red-light bed) costing roughly $125,000 USD. Athletic trainers or physical therapists administering the treatment with the aim of reducing musculoskeletal pain should apply PBM for at least 3 minutes, but no longer than 10 minutes, 3–43/week for at least 3–8 weeks. Achieving meaningful reductions in pain should theoretically allow the injured athlete to progress with their supervised training program quicker, minimizing the negative effects of chronic pain on motor control, movement kinetics, force production, adaptation, task performance, and fear avoidance. More research is needed to see if PBM can reduce the time an athlete spends away from their sport after injury. Researchers designing PBM trials in injured athletes should recruit larger cohorts, assess injury severity, use differing PBM dosing strategies, use standard performance assessments and report return-to-play assessment criteria. Examining psychological readiness for return to play alongside physical readiness also has clinical utility because it is associated with not returning to sport or returning to sport at lower levels of performance. Future research should also evaluate the effectiveness of whole body PBM equipment (red-light bed) on pain as they are commercially available, easy to operate, and do not require trained personnel to operate, reducing time burden on practitioners and increasing access to the beneficial effect of PBM (e.g., sport teams and fitness facilities). Sports scientists should continue to monitor the literature for studies addressing the above deficiencies in the current limited evidence of the effect of PBM on return to play.

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Effects of recent cannabis consumption on eye-tracking and pupillometry. Frontiers in Neuroscience, 18:1358491, 2024.

Introduction: Cannabis consumption is known to immediately affect ocular and oculomotor function; however, cannabis consumption is also known to affect it for a prolonged period of time. The purpose of this study is to identify an eye tracking or pupillometry metric which is affected after recent cannabis consumption but is not confounded by cannabis consumption history or demographic variables.

Methods: Quasi-experimental design. Participants who would consume inhalable cannabis (n = 159, mean age 31.0 years, 54% male) performed baseline neurobehavioral testing and eye-function assessments when they were sober. Eye function assessments included eye-tracking [gaze (point of visual focus), saccades (smooth movement)] and pupillometry. Participants then inhaled cannabis until they self-reported to be high and performed the same assessment again. Controls who were cannabis naïve or infrequent users (n = 30, mean age 32.6 years, 57% male) performed the same assessments without consuming cannabis in between. Results: Cannabis significantly affected several metrics of pupil dynamics and gaze. Pupil size variability was the most discriminant variable after cannabis consumption. This variable did not change in controls on repeat assessment (i.e., no learning effect), did not correlate with age, gender, race/ethnicity, or self-reported level of euphoria, but did correlate with THC concentration of cannabis inhaled.

Discussion: A novel eye-tracking metric was identified that is affected by recent cannabis consumption and is not different from non-users at baseline. A future study that assesses pupil size variability at multiple intervals over several hours and quantifies cannabis metabolites in biofluids should be performed to identify when this variable normalizes after consumption and if it correlates with blood THC levels.

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Elevated brain temperature under severe heat exposure impairs cortical motor activity and executive function. Journal of Sport and Health Science, 1392): 233-244, 2024.

Background: Excessive heat exposure can lead to hyperthermia in humans, which impairs physical performance and disrupts cognitive function. While heat is a known physiological stressor, it is unclear how severe heat stress affects brain physiology and function.

Methods: Eleven healthy participants were subjected to heat stress from prolonged exercise or warm water immersion until their rectal temperatures (Tre) attained 39.5°C, inducing exertional or passive hyperthermia, respectively. In a separate trial, blended ice was ingested before and during exercise as a cooling strategy. Data were compared to a control condition with seated rest (normothermic). Brain temperature (Tbr), cerebral perfusion, and task-based brain activity were assessed using magnetic resonance imaging techniques.

Results: Tbr in motor cortex was found to be tightly regulated at rest (37.3°C ± 0.4°C (mean ± SD)) despite fluctuations in Tre. With the development of hyperthermia, Tbr increases and dovetails with the rising Tre. Bilateral motor cortical activity was suppressed during high-intensity plantarflexion tasks, implying a reduced central motor drive in hyperthermic participants (Tre = 38.5°C ± 0.1°C). Global gray matter perfusion and regional perfusion in sensorimotor cortex were reduced with passive hyperthermia. Executive function was poorer under a passive hyperthermic state, and this could relate to compromised visual processing as indicated by the reduced activation of left lateral-occipital cortex. Conversely, ingestion of blended ice before and during exercise alleviated the rise in both Tre and Tbr and mitigated heat-related neural perturbations.

Conclusion: Severe heat exposure elevates Tbr, disrupts motor cortical activity and executive function, and this can lead to impairment of physical and cognitive performance.

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Elite athletes' and support staffs' experiences and perceptions of long-haul travel, and the self-management strategies they use. International Journal of Sports Science & Coaching, 19(2): 1-12, 2024.

Abstract - Elite athletes and their support staff are often required to travel for international competitions all over the globe, however little is known about their experiences of long-haul (LH) travel and its perceived effects on performance. The aim of this study was to explore elite athletes' and support staffs' perception of symptoms experienced following LH travel and the self-management strategies they used to mitigate these symptoms. Elite athletes and support staff (n=88), who had embarked on an LH flight (> 8 hours) in the previous 2 years for training or competition, completed a survey examining their perceptions of LH travel, symptoms experienced following the flight(s) and strategies they implemented to minimize the symptoms. Associations between symptoms experienced and travel strategies used with participant and journey characteristics were examined by Chi-squared tests. LH travel was widely perceived by participants to be disruptive to physical (86.4%) and mental performance (72.7%) and to increase the risk of illness and injury (86.4%). The most common symptoms experienced were related to fatigue and disruption to sleep. All participants implemented strategies to help mitigate the negative consequences of LH travel. Moving and stretching regularly in-flight and simple strategies for aligning the body clock to destination time were most prevalent. The study findings will allow the translation of research to better inform future guidelines that address the unique needs and priorities of elite athletes and support staff as they embark on LH travel for training and competition with a view to optimizing performance outcomes.

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Pro Football Sports Science Research Update

Epidemiology of hamstring tears in NCAA athletes: Findings from the NCAA Injury Surveillance Program between 2014/2015 and 2018/2019. Clinical Journal of Sport Medicine, Ahead of Print: 1-10, 2024.

Objective: To describe the epidemiology of hamstring tears in National Collegiate Athletic Association (NCAA) sports.

Design: Descriptive epidemiology study. Athletic trainers from NCAA schools reported injuries to the NCAA Injury Surveillance Program.

Setting: A convenience sample of NCAA hamstring tear injuries during the 2014/2015 through 2018/2019 academic years.

Patients (or Participants): NCAA student-athletes.

Independent Variables: Sport, sex, event type, season segment, injury history, and activity at the time of injury.

Main Outcome Measures: Injury counts, rates, and proportions were used.

Results: Two thousand ninety-six hamstring tears from 8 474 400 athlete-exposures (AEs) were reported (2.47 per 10 000 AEs). Rates were highest in Men's Soccer (5.97 per 10 000 AEs) and Women's Soccer (3.13 per 10 000 AEs), among all Men's and Women's sports, respectively. Competition-related rates in Men's and Women's sports were highest in 2015 to 2016 then followed a decreasing pattern across the remainder of the study period. Among sex-comparable sports, rates were higher in men's (compared with women's) Baseball/Softball, Soccer, and Track and Field. The prevalence of recurrent injuries was comparable among men's (14.8%) and women's (11.5%) sports. Time loss hamstring tears were more prevalent in Men's sports than Women's sports [injury proportion ratio 5 1.33; 95% confidence interval, (1.21, 1.47)].

Conclusions: Overall, hamstring tear rates were higher across all Men's sports compared with Women's sports. Rates across event type were comparable in several sports; and so, adjustments to practice are needed considering that practice environments are more modifiable than competitions. Indeed,

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improving hamstring tear prevention programs to reduce the burden of this injury in NCAA athletes remains critical.

Head acceleration event exposure and cognitive and functional outcomes: A comparison of multiple football seasons. Research in Sports Medicine, 32(1): 122-131, 2024.

Abstract - Athletes in contact sports are exposed to repetitive impacts as an inherent part of sport. There is concern over the accumulative effect; however, much is still unknown regarding their short-term effects. This study investigated impact accumulation and outcomes over three seasons (2015, 2017, 2019) in NCAA Football Bowl Subdivision players. Impacts were recorded using helmet accelerometers, and virtual reality testing (VR) was done across the season. Incidence rates for impacts (total; ≥25 G to <80 G; ≥80 G) all significantly differed by season ($p < 0.05$). VR scores changed across the seasons, specifically significant decreases in spatial memory ($p < 0.05$) in 2015, significant changes in balance and spatial memory ($p < 0.05$) in 2017, and no significant changes in 2019. Linear regressions predicting VR change score by impact incidence rate were nonsignificant. Monitoring exposure to impacts and changes in outcomes is useful; however, results are fluid, and many factors could indirectly have protective effects on athletes.

How mental health affects injury risk and outcomes in athletes. Sports Health, Ahead of Print: 1-8, 2024.

Context: The importance of mental health among athletes is becoming more recognized and valued. Athletes experience symptoms of depression, anxiety, and related mental health disorders at rates similar to those of the general population, but because athletes face unique cultural and environmental pressures, such disorders are exacerbated in the context of an injury. Furthermore, we review the less well-known evidence that mental health disorders in athletes are associated with an increased risk of injury. We discuss the increasing awareness of the deficiencies in mental health support for athletes,

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which has been especially highlighted during the COVID-19 pandemic as well as in prominent professional and Olympic athletes, and describe both internal and external barriers to appropriate care.

Evidence acquisition: We searched PubMed for pertinent peer-reviewed studies.

Study design: Clinical review.

Level of evidence: Level 5.

Results: There is a known psychological response to injury that can prolong recovery from musculoskeletal injury; conversely, mental health disorders in athletes are not only associated with an increased injury risk but also portend poorer outcomes subsequently, including prolonged recovery times, increased rates of injury recurrence, decreased rates of return to sport, and reduced performance upon return. Because of inherent barriers to appropriate care of athletes, including identification, stigma, and resource availability, there are currently various ongoing efforts nationally to create and implement initiatives regarding athlete mental health screening, support systems, and directed interventions to address the inextricably linked physical and mental health of athletes.

Conclusion: Athletic injury negatively impacts the mental health of athletes. Likewise, mental health can and does influence athletic performance and is also intimately tied to the risk of athletic injury, thus creating a complex cycle with inability to separate physical and mental health.

Human foot muscle strength and its association with sprint acceleration, cutting and jumping performance, and kinetics in high-level athletes. Journal of Sports Sciences, Ahead of Print: 1-12, 2024.

Abstract - The primary objective of this study was to investigate the relationship between metatarsophalangeal joint (MTPj) flexion torque and sprint acceleration, cutting and jumping performance, and kinetics. A secondary aim was to explore this relationship when MTP flexion strength was associated with other foot and lower limb neuromuscular outputs. After an initial MTPj flexion torque assessment using a custom-built dynamometer, 52 high-level athletes performed the following tasks on a force platform system: maximal sprint acceleration, 90-degree cutting, vertical and horizontal

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jumps, and foot-ankle hops. Their foot posture, foot passive stiffness and foot-ankle reactive strength were assessed using the Foot Posture Index, the Arch Height Index Measurement System and the Foot-Ankle Rebound Jump Test. Ankle plantarflexion and knee extension isometric torque were assessed using an isokinetic dynamometer. During maximal speed sprinting, multiple linear regressions suggested a major contribution of MTPj flexion torque, foot passive stiffness and foot-ankle reactive strength to explain 28% and 35% of the total variance in the effective vertical impulse and contact time. Ankle plantarflexor and quadriceps isometric torques were aggregately contributors of acceleration performance and separate contributors of cutting and jumping performance. In conclusion, MTPj flexion torque was more strongly associated with sprinting performance kinetics especially at high-speed.

Influence of the external workload on calf muscle strain injuries in professional football players: A pilot study. Sports Health, Ahead of Print: 1-8, 2024.

Background: The influence of external workload variables on the development of calf muscle strains in football players has not been previously explored.

Hypothesis: Overloaded players would have an increased risk of calf muscle strain injury.

Study design: Prospective observational study.

Level of evidence: Level 4.

Methods: A total of 41 professional football players from 1 team were monitored for 2 consecutive seasons. Total distance covered (TD), and distances covered at high-intensity running, high sprint running, low (LACC) and high (HACC) acceleration, low (LDEC) and high (HDEC) deceleration, and at high metabolic load distance (HMLD) were monitored with GPS units. Accumulated players' external workload in the week before injury was compared with the weekly mean value of the 6 weeks before injury occurred for each player.

Results: Ten players (24.3%) suffered 16 calf muscle strain injuries (3.1 injuries per 1000 hours of match play; 0.5 injuries per 1000 hours of training exposure). Players with a calf muscle injury were older ($p = 0.03$), with higher body weight ($p = 0.01$) and height ($p = 0.03$). Injured players displayed substantially higher total training volume ($p < 0.01$), TD ($p < 0.01$), LACC ($p < 0.01$), LDEC ($p < 0.01$), HACC ($p < 0.01$),

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HDEC ($p < 0.01$), and HMLD ($p = 0.03$) in the week before injury, in comparison with the mean values of these variables in the 6 weeks before injury.

Conclusion: A week with a higher-than-habitual external workload might increase the risk of calf muscle strain injury in professional football players. Calf muscle injuries were preceded by a week with unusually high workloads associated with accelerating and decelerating distances and higher training volumes.

Clinical relevance: Monitoring external workload indicators may be helpful in determine players with a higher risk of calf muscle strain injury due to excessive workload during training/competition.

Knee cartilage injuries in football players: Clinical outcomes and return to sport after surgical treatment: A systematic review of the literature. Cartilage, Ahead of Print: 1-15, 2024.

Objective: To systematically review the literature and analyze clinical outcomes and return-to-sport after surgical management of cartilage injuries in football players.

Design: A systematic literature review was performed in August 2023 on PubMed, WebOfScience, and Cochrane Library to collect studies on surgical strategies for cartilage lesions in football players. Methodological quality and risk of bias were assessed with the modified Coleman Methodology score and RoB2 and RoBANS2 tools.

Results: Fifteen studies on 409 football players (86% men, 14% women) were included: nine prospective and two retrospective case series, one randomized controlled trial, one prospective comparative study, one case report, and one survey. Bone marrow stimulation (BMS) techniques were the most documented. The lesion size influenced the treatment choice: debridement was used for small lesions (1.1 cm2), BMS, osteochondral autograft transplantation (OAT), matrix-assisted autologous chondrocytes transplantation (MACT), and scaffold-augmented BMS for small/mid-size lesions (2.2-3.0 cm2), and autologous chondrocytes implantation (ACI) for larger lesions (5.8 cm2). The surgical options yielded different results in terms of clinical outcome and return-to-sport, with fastest recovery for

debridement and scaffold-augmented BMS. The current evidence is limited with large methodological quality variation (modified Coleman Methodology score 43.5/100) and a high risk of bias.

Conclusions: Decision-making in cartilage injuries seems to privilege early return-to-sport, making debridement and microfractures the most used techniques. The lesion size influences the treatment choice. However, the current evidence is limited. Further studies are needed to confirm these findings and establish a case-based approach to treat cartilage injuries in football players based on the specific patient and lesion characteristics and the treatments' potential in terms of both return-to-sport and long-term results.

Level of evidence: Systematic review, level IV.

Long-term outcomes of multiligament knee injuries in American football players. The American Journal of Sports Medicine, Ahead of Print: 1-9, 2024.

Background: Long-term outcomes for isolated anterior cruciate ligament (ACL) reconstructions in competitive American football athletes are well reported in the literature, but little data currently exist regarding multiligament knee injury (MLKI) reconstruction outcomes.

Purpose: To examine patient-reported and return-to-sport outcomes of competitive American football athletes who underwent primary, single-staged, multiligament knee reconstruction.

Study design: Case series; Level of evidence, 4.

Methods: We identified patients from our institution's prospectively collected data repository between 2001 and 2020 who underwent single-staged surgical reconstruction of an MLKI sustained during competitive participation in American football. We assessed patient-reported outcomes at a minimum of 2 years after surgery using the International Knee Documentation Committee (IKDC) Subjective Knee Form and questions regarding surgical satisfaction and return to sport. Successful return to sport was defined as a return to preinjury level of competition. We summarized all outcome data and compared outcomes between 2-ligament and >2-ligament groups and between ACL-only MLKI injury and bicruciate MLKI injury groups using independent t test for IKDC scores and chi-square test for return to sport.

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Additionally, we evaluated predictors of postoperative IKDC scores using linear regression and predictors of return to sport using logistic regression.

Results: Outcome data were successfully collected for 53 of 73 total eligible patients (73%; mean follow-up time, 7.7 ± 4.0 years; all male; mean age at surgery, 18.1 ± 2.7 years). The mean postoperative IKDC score was 84 ± 16. The most common level of preinjury competition was high school (n = 36; 68%), followed by college (n = 10; 19%). Seven patients did not return to sport competition at any level due to limitations from their knee surgery, and 82% of patients that attempted to return to preinjury level of sport were able to do so. A total of 50 patients (94%) were satisfied or very satisfied with their surgical outcome. The 2-ligament (n = 39) and >2-ligament (n = 14) groups did not significantly differ in IKDC scores (P = .96) or proportions with successful return to sport (P = .77). Similarly, the ACL-MLKI injury (n = 39) and bicruciate MLKI injury (n = 14) groups did not significantly differ in IKDC scores (P = .89) or proportions with successful return to sport (P = .77). Age and body mass index were not significantly associated with IKDC scores or successful return to sport at follow-up (all P > .05).

Conclusion: This study may represent the largest cohort of competitive American football athletes evaluated for longitudinal outcomes after multiligament knee reconstruction. Despite the severity of these injuries, we found good knee-related function and that the large majority of athletes who attempted to return to sport were successful. The majority of athletes (94%) were satisfied with their operative treatment.

Maximum speed exposures in Australian Rules football: Do methods matter? Science and Medicine in Football, 8(3): 287-290, 2024.

Abstract - This study compared weekly near-to-maximum speed exposure data when determined via different methods and criteria. Forty-seven professional Australian rules football (AF) players undertook pre-season sprint assessments (PSSA) via 3 × 50 m maximal efforts using 10 Hz GPS over two consecutive seasons. The same technology was used continually during the in-season to identify maximum speeds attained in training and matches. Weekly near-to-maximal speed (MS) exposure counts were aggregated for speeds ≥80%, ≥85%, ≥90% and ≥95% of their individual maximum for both determination approaches. Weekly near-to-MS exposures was lower ($p < 0.0001$) when determined from in-season

monitoring for ≥80% (-1.26; CI: -1.58 to -0.93), ≥85% (-0.78; CI: -0.97 to -0.59), ≥90% (-0.42; CI: -0.53 to -0.32), and ≥95% (-0.09; CI: -0.12 to -0.06) versus PSSA, with no effect of playing position (P ≥ 0.161). Although ≥80% and ≥85% near-to-maximum speed exposure data was meaningfully influenced by the determination method, the effect was somewhat trivial at higher speed criteria (≥90% and ≥95%) often considered important for performance gains and injury risk reduction purposes. Maximum speed determination methods therefore may be used interchangeably, and discrete sprint assessments may not be necessary for this purpose.

NFL quarterbacks with ulnar collateral ligament injuries have high return to play rates, but older players have inferior post-injury performance. Arthroscopy, Sports Medicine, and Rehabilitation, Ahead of Print: 1-30, 2024.

Purpose: The purpose of this study was to characterize the epidemiology, mechanism (contact versus non-contact), management, and return to play times for NFL quarterbacks who experienced UCL injuries.

Methods: Using the Pro Sports Transactions Archive, NFL quarterbacks who sustained UCL injuries between 1991 and 2023 were identified. Age at time of injury, mechanism of injury (contact vs non-contact), management strategy, and return to play time (RTP) were recorded. Additionally, player performance metrics including games played, quarterback rating, completion percentage, touchdowns, and interceptions, were examined for the season before injury through the second season post-injury. Player performance analysis was assessed using principal component analysis (PCA), which is a dimensionality reduction statistical method that compresses the several performance metrics into a single value, the first principal component (PC1).

Results: A total of 21 injuries in 20 players were identified. The mean age of quarterbacks at time of injury was 28.7 years (SD = ± 4.6 years). Ten injuries occurred during contact with another player and 11 injuries were non-contact. The majority (13) of UCL injuries were managed non-operatively. Sixteen players achieved RTP, four of which underwent repair or reconstruction (UCLR). Overall, the mean RTP time was 165.6 days (SD = ± 178.8 days), but players who underwent UCLR had a longer average RTP time (359.0 days v. 98.25 days, p = 0.014). Injuries sustained after 2006 were associated with improved

post injury performance (p = 0.041), but older age at time of injury was associated with diminished post injury performance (p =0.048).

Conclusions: NFL quarterbacks sustain more non-contact UCL injuries and are undergoing UCLR at higher rates than previously reported. Although RTP rates are high and players demonstrate improved post injury performance for injuries sustained after 2006, older age at the time of injury was associated with worse post injury performance.

Overview of treatment options for mild traumatic brain injury: A literature review. Cureus, 16(4): e59021, 2024.

Abstract - The incidence and prevalence of concussion, a type of mild traumatic brain injury (mTBI), have steadily increased among athletes, both students and professionals, across a wide variety of sports, including, but not limited to, swimming, tennis, football, and boxing. Recent data have demonstrated that sports are one of the leading causes of concussions among student athletes. While the exact mechanism of concussion onset has yet to be fully elucidated, data suggest that the pathophysiology involves rotational acceleration and deceleration of the brain, leading to axon tearing and disturbance in the metabolic cascade of glucose. Concussive events can have debilitating effects on an athlete, including chronic traumatic encephalopathy (gradual degeneration of brain tissue) that is related to personality changes, emotional disorders, and even dementia. Common symptoms associated with concussion include dizziness, nausea, vomiting, and headaches. The physical assessment consists of a combination of tools involving the mental status examination, vital signs, cervical spine exam, eye exam, and neurological testing. The use of osteopathic manipulative medicine (OMM), pharmacotherapy, hyperbaric oxygen therapy (HBOT), aerobic exercise, balance, and/or vestibular therapy are many common treatment approaches for concussion and post-concussion sequelae. This literature review aims to provide insight into concussions, the current treatment options available, and the new developments in concussions per the Amsterdam 2022 International Consensus Statement on Concussion in Sport published in 2023.

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Physician perceptions of cannabidiol (CBD) and cannabis in sports medicine and performance. Translational Sports Medicine, 11:2023:8824466, 2023.

Objectives: There is growing evidence regarding cannabinoid use in sports medicine and performance, especially cannabidiol (CBD). This study aims to determine if sports medicine physicians are recommending cannabinoids for therapeutic purposes, as well as analyze perceptions of cannabinoids within sports medicine and performance.

Methods: Physician members of the American Medical Society for Sports Medicine (AMSSM) completed an anonymous survey on demographics, CBD and Cannabis recommendations, as well as attitudes toward cannabinoid products within sports medicine. Factors associated with CBD and cannabis recommendations as well as perceptual differences were found using multivariate regression modelling.

Results: Responses from 333 physicians were recorded. The following groups were less likely to agree with allowing cannabis for recreational purposes: female gender (coeff. = 0.79 (0.33-1.25), $p=0.001$), increasing age (coeff. = 0.04 (0.02, 0.07), $p < 0.001$), and rural respondents (compared to baseline urban, coeff. = 1.16 (0.36, 1.95), $p=0.004$). Similarly, these three factors were associated with a higher likelihood of disagreeing with WADA removing cannabis from the prohibited substance list and with the NCAA allowing CBD use by collegiate athletes ($p \leq 0.045$). CBD was less likely to be recommended by pediatricians, rural physicians, and academic physicians ($p \leq 0.030$). Male physicians and younger physicians were less likely to identify cannabis as performance-enhancing ($p \leq 0.042$).

Conclusions: Sports medicine physicians have varying views on cannabinoids. While sports medicine physicians generally have favorable attitudes toward CBD and cannabis, these perceptions appear to be significantly affected by age, practice type, and gender.

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Predictors of early return to sport after surgical repair of proximal hamstring complex injuries in professional athletes: A prospective study. The American Journal of Sports Medicine, 52(4): 1005-1013, 2024.

Background: Proximal hamstring complex injury (PHCI) is a common injury among professional athletes, particularly those participating in pivot contact sports. Previous studies have suggested that surgery can be effective in restoring function and allowing athletes to return to sport (RTS), but the factors influencing successful RTS have been less clear.

Purpose: To assess RTS capabilities after surgical treatment of PHCI in professional athletes and to identify favorable predictors of RTS.

Study design: Case series; Level of evidence, 4.

Methods: This study (2002-2022) was conducted on professional athletes who underwent surgical treatment for PHCI at a sports surgery center. The primary outcome of the study was the RTS capability, evaluated based on the rate of athletes' return to their preinjury level of competition, time delay to RTS, and quality of RTS as measured using their level of performance and progression of scores on activity scales such as the Tegner Activity Scale (TAS) and University of California, Los Angeles (UCLA), scale. "Maintained performance" was defined as athletes returning to the same preinjury activity level (per the TAS and UCLA scale) and perceiving themselves to have maintained their performance. Secondary outcomes covered the potential RTS predictors and complication rate. The study distinguished 2 types of PHCI: proximal hamstring tendon avulsion injury (proximal rupture with empty footprint, or having a "positive dropped ice cream sign") and complete proximal hamstring free tendon rupture (PHTR; proximal rupture without empty footprint, or having a "negative dropped ice cream sign").

Results: The study examined 64 professional athletes (mean age, 27.3 years; 82.8% male) undergoing surgery for PHCI. The RTS rate was 98.4%, with 78.1% of the athletes returning to their preinjury level of competition at 6.2 months (SD, 2.5 months). Twelve (19%) patients had returned to sport at an inferior level of competition, and 2 (3.1%) were unable to continue in their preinjury sport. Subgroup analysis revealed variation in RTS based on sport type, with the highest rate of return to preinjury performance found in athletes in handball and sports with splits (fencing, squash, and escalade; 100%) and soccer (95.2%). In the univariate analysis, male sex (hazard ratio [HR], 4.05; 95% CI, 1.45-11.3; P = .008), higher

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preinjury TAS score (HR, 1.27; 95% CI, 1.06-1.52; P = .011), injury involving the semimembranosus (HR, 4.84; 95% CI, 2.31-10.2; P < .001) or conjoint tendon (HR, 3.12; 95% CI, 1.55-6.25; P = .001), and PHTR (HR, 7.77; 95% CI, 3.54-17.0; P < .001) were significantly associated with a better postoperative level of competition. Multivariate analysis identified 3 favorable predictors of RTS with HRs of 2.91 (95% CI, 1.01-8.35; P = .047) for male sex, 3.86 (95% CI, 1.78-8.37; P < .001) for isolated semimembranosus injury, and 5.18 (95% CI, 2.24-12.0; P < .001) for PHTR. The complication rate was 4.7%.

Conclusion: Favorable predictors of early RTS were male sex, isolated semimembranosus injury, and PHTR injuries.

Preventing exertional heat stroke in football: Time for a paradigm shift. Sports Health, Ahead of Print: 1-7, 2024.

Context: Among American sports, football has the highest incidence of exertional heat stroke (EHS), despite decades of prevention strategies. Based on recent reports, 100% of high school and college EHS football fatalities occur during conditioning sessions. Linemen are the at-risk population, constituting 97% of football EHS deaths. Linemen heat up faster and cool down slower than other players.

Evidence acquisition: Case series were identified from organized, supervised football at the youth, high school, and collegiate levels and compiled in the National Registry of Catastrophic Sports Injuries. Sources for event occurrence were media reports and newspaper clippings, autopsy reports, certificates of death, school-sponsored investigations, and published medical literature. Articles were identified through PubMed with search terms "football," "exertional heat stroke," and "prevention."

Study design: Clinical review.

Level of evidence: Level 5.

Results: Football EHS is tied to (1) high-intensity drills and conditioning that is not specific to individual player positions, (2) physical exertion as punishment; (3) failure to modify physical activity for high heat and humidity, (4) failure to recognize early signs and symptoms of EHS, and (5) death when cooling is delayed.

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Conclusion: To prevent football EHS, (1) all training and conditioning should be position specific; (2) physical activity should be modified per the heat load; (3) understand that some players have a "do-or-die" mentality that supersedes their personal safety; (4) never use physical exertion as punishment; (5) eliminate conditioning tests, serial sprints, and any reckless drills that are inappropriate for linemen; and (6) consider air-conditioned venues for linemen during hot practices. To prevent EHS, train linemen based on game demands.

Reasons for not returning to pre-injury sport level after ACL-reconstruction. International Journal of Sports Medicine, Ahead of Print: 1-7, 2024.

Abstract - The study aimed to identify athlete-reported reasons for not returning to pre-injury sports level after anterior cruciate ligament reconstruction (ACLR) and to identify the factors associated with these reasons. Ninety-one athletes with 2 years post-ACLR indicated whether or not they had returned to their pre-injury sport level (same frequency, duration, and intensity). Athletes who did not return were asked to provide the reasons. Athletes' characteristics and injury-related factors were used to determine factors associated with the reasons for not returning. Only nine athletes (10%) returned to pre-injury sport level after ACLR. The most common reasons for not returning were lack of confidence or concerns about re-injury (48.8%), followed by continued post-surgical impairments in the reconstructed knee (39%). Having episodes of the knee giving way after ACLR was the only significant predictor of post-surgical impairments (48.8%; OR=8.3, 95%CI=2.48-27.42, p=0.001). Lack of confidence, concerns about re-injury, or post-surgical impairments in the reconstructed knee were the most frequently reported reasons for not returning to pre-injury sports level with 2 years post-ACLR. Reported dynamic knee instability was the only factor associated with ongoing post-surgical knee impairments after ACLR. Rehabilitation programs should address athletes' psychological responses and resolve knee impairments to optimize return to pre-injury sport level after ACLR.

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Return to the pre-injury level of sport after anterior cruciate ligament reconstruction: A practical review with medical recommendations. International Journal of Sports Medicine, Ahead of Print: 1-17, 2024.

Abstract - Returning to sport after anterior cruciate ligament reconstruction (ACLR) can be a challenging and complex process for the athlete, with the rate of return to the pre-injury level of sport observed to be less than athlete expectations. Of the athletes that do return to sport (RTS), knee re-injury rates remain high, and multiple studies have observed impaired athletic performance upon RTS after ACLR as well as reduced playing time, productivity, and career lengths. To mitigate re-injury and improve RTS outcomes, multiple RTS after ACLR consensus statements/clinical practice guidelines have recommended objective RTS testing criteria to be met prior to medical clearance for unrestricted sports participation. While the achievement of RTS testing criteria can improve RTS rates after ACLR, current criteria do not appear valid for predicting safe RTS. Therefore, there is a need to review the various factors related to the successful return to the pre-injury level of sport after ACLR, clarify the utility of objective performance testing and RTS criteria, further discuss safe RTS decision-making as well as present strategies to reduce the risk of ACL injury/re-injury upon RTS. This article provides a practical review of the current RTS after ACLR literature, as well as makes medical recommendations for rehabilitation and RTS decision-making after ACLR.

Skimmed, lactose-free milk ingestion postexercise: Rehydration effectiveness and gastrointestinal disturbances vs water and a sports drink in physically active people. International Journal of Sport Nutrition and Exercise Metabolism, Ahead of Print: 1-9, 2024.

Abstract - Postexercise hydration is fundamental to replace fluid loss from sweat. This study evaluated rehydration and gastrointestinal (GI) symptoms for each of three beverages: water (W), sports drink (SD), and skimmed, lactose-free milk (SLM) after moderate intensity cycling in the heat. Sixteen college students completed three exercise sessions each to lose ≈2% of their body mass. They drank 150% of body mass loss of the drink assigned in randomized order; net fluid balance, diuresis, and GI symptoms

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were measured and followed up for 3 hr after completion of fluid intake. SLM showed higher fluid retention (~69%) versus W (~40%; $p < .001$); SD (~56%) was not different from SLM or W ($p > .05$). Net fluid balance was higher for SLM (−0.26 kg) and SD (−0.42 kg) than W (−0.67 kg) after 3 hr ($p < .001$), resulting from a significantly lower diuresis with SLM. Reported GI disturbances were mild and showed no difference among drinks ($p > .05$) despite ingestion of W (1,992 ± 425 ml), SD (1,999 ± 429 ml), and SLM (1,993 ± 426 ml) in 90 min. In conclusion, SLM was more effective than W for postexercise rehydration, showing greater fluid retention for the 3-hr follow-up and presenting with low-intensity GI symptoms similar to those with W and SD. These results confirm that SLM is an effective option for hydration after exercise in the heat.

TeamWellX: Leveraging genetic algorithms to optimize team health and well-being toward sustainable game development and strategy. SoftwareX, 101635, 2024.

Abstract - Building a well-rounded roster is crucial in sports, as it directly impacts a team's long-term success. However, the process can be complex, with various factors to consider, such as team dynamics, player skills, and financial limitations. This study aims to develop effective strategies for constructing teams in different sports, starting with American Football in the NFL and potentially expanding to other similar disciplines. In this paper, we propose an innovative tool that leverages genetic algorithm, signifying a method that balances between overall team positional strength, fosters collective team success, and encourages team's long-standing well-being. This data-driven approach also presents a generalizable solution that can be employed across similar team sports, transforming the pattern of roster optimization strategies and team's performance augment.

Pro Football Sports Science Research Update

The effect of variations in knee and hip angles on electromyographic activity of the hamstrings and related muscles during the Nordic hamstring exercise. International Journal of Sports Physical Therapy, 19(3): 275-283, 2024.

Background: The benefit of performing the Nordic Hamstring Exercise (NHE) on an inclined board has been described, however, isometric hamstring activation in different knee and hip angles has not yet been thoroughly explored.

Purpose: This study investigated the effect of variations in knee and hip angles during the isometric performance of the NHE on electromyographic activity of the hamstring muscles.

Study design: Crossover study.

Methods: Thirteen male volunteers performed isometric contractions during the NHE with the knee (30°, 50°, 60°) and the hip (0°, 30°, and 45°) in various angles of flexion on a leg support platform which was inclined at 30°. An electrical goniometer was used to monitor the knee and hip joint angles during 5-s isometric contractions. A multivariate analysis of variance with repeated measures was used to compare normalized electromyographic values of each muscle across different knee and hip angles, followed by pairwise comparisons.

Results: The electromyographic activity of the biceps femoris, semitendinosus, and semimembranosus at a knee angle of 30° and hip angle of 0° were significantly higher than those observed with a knee angle of 50° and hip angle of 0°, or a knee angle of 60° and hip angle of 0° ($p<0.05$). The electromyographic activity of the semimembranosus at a knee angle of 60° and hip angle of 45° was significantly higher than values obtained with knee and hip angles of 60° and 0°, respectively ($p<0.05$).

Conclusions: The results indicate that using a knee flexion of 30° and a hip flexion of 0°, while isometrically performing the NHE on a platform inclined at 30°, may optimize electromyographic activity of the hamstrings.

Level of evidence: 3.

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Pro Football Sports Science Research Update

Thoracolumbar fascia and lumbar muscle stiffness in athletes with a history of hamstring injury. Journal of Sports Science and Medicine, 23: 436-444, 2024.

Abstract - The purpose of this study was to examine the differences in thoracolumbar fascia (TLF) and lumbar muscle modulus in individuals with and without hamstring injury using shear wave elastography (SWE). Thirteen male soccer players without a previous ham string injury and eleven players with a history of hamstring injury performed passive and active (submaximal) knee flexion efforts from 0°, 45° and 90° angle of knee flexion as well as an active prone trunk extension test. The elastic modulus of the TLF, the erector spinae (ES) and the multifidus (MF) was measured using ultrasound SWE simultaneously with the surface electromyography (EMG) signal of the ES and MF. The TLF SWE modulus was significantly ($p < 0.05$) higher in the injured group (range: 29.86 ± 8.58 to 66.57 ± 11.71 kPa) than in the uninjured group (range: 17.47 ± 9.37 to 47.03 ± 16.04 kPa). The ES and MF modulus ranged from 14.97 ± 4.10 to 66.57 ± 11.71 kPa in the injured group and it was significantly ($p < .05$) greater compared to the uninjured group (range: 11.65 ± 5.99 to 40.49 ± 12.35 kPa). TLF modulus was greater than ES and MF modulus ($p < 0.05$). Active modulus was greater during the prone trunk extension test compared to the knee flexion tests and it was greater in the knee flex ion test at 0° than at 90° ($p < 0.05$). The muscle EMG was greater in the injured compared to the uninjured group in the passive tests only ($p < 0.05$). SWE modulus of the TLF and ES and MF was greater in soccer players with previous hamstring injury than un injured players. Further research could establish whether exercises that target the paraspinal muscles and the lumbar fascia can assist in preventing individuals with a history of hamstring injury from sustaining a new injury.

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Pro Football Sports Science Research Update

Validity of the Hawkin Dynamics wireless dual force platform system against a piezoelectric laboratory grade system for vertical countermovement jump variables. Journal of Strength and Conditioning Research, 38(6): 1144-1148, 2024.

Abstract - The aim of this study was to determine the criterion validity of the Hawkin Dynamics (HD) wireless dual force platform system for assessing vertical countermovement jump (CMJ) variables, compared with those derived from a Kistler piezoelectric laboratory grade force platform system. During a single testing session, HD force platforms were placed directly on top of 2 adjacent Kistler force platforms to simultaneously collect vertical ground reaction forces produced by 2 male recreational soccer players (age: 29.0 ± 2.8 years, height: 1.79 ± 0.01 m, mass: 85.6 ± 4.7 kg) that performed 25 vertical CMJs each. Sixteen vertical CMJ variables pertaining to jump height (JH), flight time (FT), time-to-take off (TTT), countermovement depth, body weight (BW), propulsive and braking mean, and peak powers, forces, and impulses were compared between systems. Fixed bias was observed for 6 of 16 variables (peak and mean braking power, mean propulsion force, TTT, FT, and BW), while proportional bias was present for 10 of 16 variables (peak and mean propulsive and braking force, TTT, FT, peak and mean braking power, mean propulsive power, and BW). For all variables regardless of fixed or proportional bias, percentage differences were ≤3.4% between force platform systems, with near perfect to perfect correlations (r or ρ = 0.977–1.000) observed for 15 of 16 variables. The HD dual wireless force platform system can be considered a valid alternative to a piezoelectric laboratory grade force platform system for the collection of vertical CMJ variables, particularly outcome (i.e., JH, reactive strength index modified) and strategy variables (countermovement depth).

Practical Applications - The HD dual wireless force platform system can be considered a valid alternative to piezoelectric laboratory grade force platform system (i.e., Kistler) for collecting vertical CMJ variables, particularly outcome and strategy metrics. As such, this cheaper, portable dual wireless force platform system can be used confidently by practitioners and researchers to evaluate the lower-limb impulsive properties of athletes via the vertical CMJ. The data collected during such tests can then be accurately used to help individualize training, for talent identification purposes, to assess neuromuscular readiness, for injury risk mitigation profiling, and making return to play from injury decisions. However, practitioners are still advised to standardize the force platform systems between testing occasions.

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